

MATH-1014 T10A

Tutorial 4

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Today's Roadmap

1 Arc Length and Surface of Revolution

2 Trigonometric Integrals

Arc Length: Core Formulas

Arc Length Formula (in terms of x)

If $y = f(x)$ on $[a, b]$ and $f'(x)$ is continuous, then

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx.$$

Arc Length Formula (in terms of y)

If $x = g(y)$ on $[c, d]$ and $g'(y)$ is continuous, then

$$L = \int_c^d \sqrt{1 + (g'(y))^2} dy.$$

Tip: Choose the form that makes the derivative and integrand simpler.

Problem 1: Arc Length

Problem 1. Find the arc length of

$$y = \ln(\cos x), \quad 0 \leq x \leq \frac{\pi}{3}.$$

Hint

Differentiate first. A nice simplification occurs in

$$\sqrt{1 + (y')^2}.$$

Solution 1

1. Differentiate:

$$y = \ln(\cos x) \implies y' = \frac{-\sin x}{\cos x} = -\tan x.$$

2. Arc-length integrand:

$$\sqrt{1 + (y')^2} = \sqrt{1 + \tan^2 x} = \sqrt{\sec^2 x} = \sec x \quad (0 \leq x \leq \frac{\pi}{3}).$$

3. Integrate:

$$L = \int_0^{\pi/3} \sec x \, dx = [\ln |\sec x + \tan x|]_0^{\pi/3}.$$

$$L = \ln(2 + \sqrt{3}) - \ln(1) = \boxed{\ln(2 + \sqrt{3})}.$$

Surface Area of Revolution: Formula Summary

About the x -axis

For $y = f(x) \geq 0$ on $[a, b]$:

$$S = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx$$

For $x = g(y)$ on $[c, d]$:

$$S = \int_c^d 2\pi y \sqrt{1 + (g'(y))^2} dy$$

About the y -axis

For $y = f(x)$ on $[a, b]$:

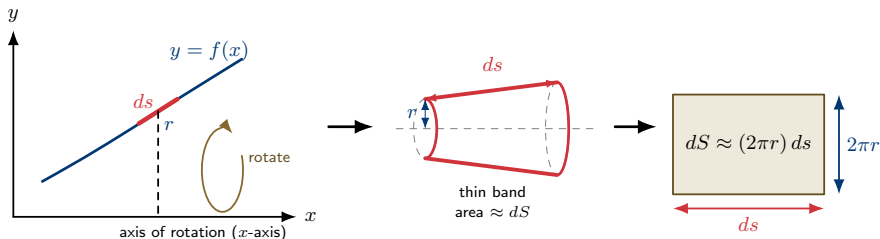
$$S = \int_a^b 2\pi x \sqrt{1 + (f'(x))^2} dx$$

For $x = g(y) \geq 0$ on $[c, d]$:

$$S = \int_c^d 2\pi g(y) \sqrt{1 + (g'(y))^2} dy$$

Reminder: Radius = distance to axis of rotation.

Geometric Idea Behind the Surface Area Formula



Key idea: A tiny arc segment of length ds rotates into a thin band, and
 area of band $\approx (\text{circumference})(\text{slanted length}) = (2\pi r) ds$.

Problem 2: Surface Area

Problem 2. Find the exact surface area generated by rotating

$$x = 3 + y^2, \quad 1 \leq y \leq 3$$

about the x -axis.

Use the formula for $x = g(y)$ about the x -axis:

$$S = \int_c^d 2\pi y \sqrt{1 + (g'(y))^2} dy.$$

Solution 2

Here $x = g(y) = 3 + y^2$, so

$$g'(y) = 2y.$$

1. Set up the integral:

$$S = \int_1^3 2\pi y \sqrt{1 + (2y)^2} dy = 2\pi \int_1^3 y \sqrt{1 + 4y^2} dy.$$

2. Substitution: Let

$$u = 1 + 4y^2, \quad du = 8y dy \quad \implies \quad y dy = \frac{du}{8}.$$

When $y = 1$, $u = 5$; when $y = 3$, $u = 37$.

$$S = 2\pi \cdot \frac{1}{8} \int_5^{37} u^{1/2} du = \frac{\pi}{4} \int_5^{37} u^{1/2} du.$$

3. Evaluate:

$$S = \frac{\pi}{4} \left[\frac{2}{3} u^{3/2} \right]_5^{37} = \boxed{\frac{\pi}{6} \left(37^{3/2} - 5^{3/2} \right)}.$$

Problem 3: Surface Area with Exponential

Problem 3. Calculate the surface area formed by rotating

$$y = 2e^{-3x}, \quad 0 \leq x \leq 2$$

about the x -axis.

Hint

$$S = \int_0^2 2\pi y \sqrt{1 + (y'(x))^2} dx$$

and then use a substitution involving e^{-3x} .

Solution 3

1. Set up the surface-area integral:

$$S = \int_0^2 2\pi(2e^{-3x})\sqrt{1+36e^{-6x}} dx = 4\pi \int_0^2 e^{-3x}\sqrt{1+36e^{-6x}} dx.$$

2. Substitution: Let

$$u = 6e^{-3x} \implies du = -18e^{-3x} dx.$$

So $e^{-3x} dx = -\frac{1}{18}du$, and the bounds become

$$x = 0 \Rightarrow u = 6, \quad x = 2 \Rightarrow u = 6e^{-6}.$$

$$S = \frac{2\pi}{9} \int_{6e^{-6}}^6 \sqrt{1+u^2} du.$$

3. Exact answer: Using

$$\int \sqrt{1+u^2} du = \frac{1}{2} \left(u\sqrt{1+u^2} + \ln(u + \sqrt{1+u^2}) \right) + C,$$

we obtain

$$S = \frac{\pi}{9} \left[u\sqrt{1+u^2} + \ln(u + \sqrt{1+u^2}) \right]_{u=6e^{-6}}^{u=6}.$$

Review: Sum & Difference Formulas

Basic Identities

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}, \quad \tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

Review: Product-to-Sum Identities

Useful for Integration

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

These identities are especially useful when powers are not convenient.

Problem 4: A Clever Product-to-Sum Application

Problem 4. Evaluate

$$I = \int_0^{\pi/2} \sin(3x) \cos(5x) dx.$$

Hint

This is not a power integral, so the $\sin^m x \cos^n x$ strategy is not convenient. Use

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)].$$

Solution 4

Use

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)].$$

With $A = 3x$, $B = 5x$, we get

$$\sin(3x) \cos(5x) = \frac{1}{2} [\sin(8x) + \sin(-2x)] = \frac{1}{2} [\sin(8x) - \sin(2x)].$$

Therefore,

$$I = \frac{1}{2} \int_0^{\pi/2} (\sin 8x - \sin 2x) dx = \frac{1}{2} \left[-\frac{\cos 8x}{8} + \frac{\cos 2x}{2} \right]_0^{\pi/2}.$$

Evaluate the endpoints:

$$\cos(4\pi) = 1, \quad \cos(\pi) = -1, \quad \cos(0) = 1.$$

So

$$I = \frac{1}{2} \left[\left(-\frac{1}{8} - \frac{1}{2} \right) - \left(-\frac{1}{8} + \frac{1}{2} \right) \right] = \frac{1}{2}(-1) = \boxed{-\frac{1}{2}}.$$

Review: Basic Trigonometric Anti-derivatives

Core Formulas

$$\int \sin x \, dx = -\cos x + C, \quad \int \cos x \, dx = \sin x + C$$

$$\int \tan x \, dx = -\ln |\cos x| + C$$

$$\int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

$$\int \csc x \, dx = -\ln |\csc x + \cot x| + C$$

Strategy: Integrals of $\sin^m x \cos^n x$

Consider

$$\int \sin^m x \cos^n x dx.$$

Case 1: m odd

If $m = 2k + 1$, split off one $\sin x$:

$$\sin^m x = (1 - \cos^2 x)^k \sin x,$$

then use

$$u = \cos x.$$

Case 2: n odd

If $n = 2k + 1$, split off one $\cos x$:

$$\cos^n x = (1 - \sin^2 x)^k \cos x,$$

then use

$$u = \sin x.$$

Strategy: Integrals of $\sin^m x \cos^n x$

Consider

$$\int \sin^m x \cos^n x dx.$$

Case 3: both m, n even

Use half-angle identities repeatedly:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

Then reduce the powers step by step.

Quick Decision Rule

- $\sin$ odd $\Rightarrow$ save one $\sin x$, use $u = \cos x$
- $\cos$ odd $\Rightarrow$ save one $\cos x$, use $u = \sin x$
- both even $\Rightarrow$ half-angle identities

Problem 5: Power of Sine/Cosine with πx

Problem 5. Evaluate

$$\int \sin^2(\pi x) \cos^5(\pi x) dx.$$

Hint: $\cos^5(\pi x)$ has an odd power. Split off one factor $\cos(\pi x)$, then use

$$u = \sin(\pi x).$$

Solution 5

$$I = \int \sin^2(\pi x) \cos^5(\pi x) dx = \int \sin^2(\pi x) (1 - \sin^2(\pi x))^2 \cos(\pi x) dx.$$

Let

$$u = \sin(\pi x), \quad du = \pi \cos(\pi x) dx \quad \implies \quad \cos(\pi x) dx = \frac{du}{\pi}.$$

Then

$$I = \frac{1}{\pi} \int u^2 (1 - u^2)^2 du = \frac{1}{\pi} \int (u^2 - 2u^4 + u^6) du.$$

$$I = \frac{1}{\pi} \left(\frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7} \right) + C.$$

Substitute back $u = \sin(\pi x)$:

$$I = \frac{1}{\pi} \left(\frac{\sin^3(\pi x)}{3} - \frac{2\sin^5(\pi x)}{5} + \frac{\sin^7(\pi x)}{7} \right) + C.$$

Problem 6: Composite Angle and Root

Problem 6. Evaluate

$$\int \frac{\sin^3(\sqrt{x})}{\sqrt{x}} dx.$$

Hint

First remove $\sqrt{x}$ from the denominator by letting $t = \sqrt{x}$. Then apply the odd-power sine strategy.

Solution 6

Let

$$t = \sqrt{x} \implies x = t^2, \quad dx = 2t \, dt.$$

Then

$$\int \frac{\sin^3(\sqrt{x})}{\sqrt{x}} dx = \int \frac{\sin^3 t}{t} (2t \, dt) = 2 \int \sin^3 t \, dt.$$

Use

$$\sin^3 t = \sin t(1 - \cos^2 t).$$

Let $u = \cos t$, so $du = -\sin t \, dt$. Then

$$\int \sin^3 t \, dt = \int \sin t(1 - \cos^2 t) \, dt = -\int (1 - u^2) \, du = -u + \frac{u^3}{3} + C.$$

Hence

$$2 \int \sin^3 t \, dt = -2 \cos t + \frac{2}{3} \cos^3 t + C.$$

Substitute back $t = \sqrt{x}$:

$$\boxed{\int \frac{\sin^3(\sqrt{x})}{\sqrt{x}} dx = -2 \cos(\sqrt{x}) + \frac{2}{3} \cos^3(\sqrt{x}) + C.}$$

Problem 7: Even-Even Powers

Problem 7. Evaluate

$$\int \sin^2 x \cos^2 x \, dx.$$

Use half-angle identities:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

Solution 7

$$\sin^2 x \cos^2 x = \frac{1 - \cos^2 2x}{4} = \frac{\sin^2 2x}{4}.$$

Now use $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$:

$$\sin^2 x \cos^2 x = \frac{1}{4} \cdot \frac{1 - \cos 4x}{2} = \frac{1 - \cos 4x}{8}.$$

Therefore

$$\int \sin^2 x \cos^2 x \, dx = \int \frac{1 - \cos 4x}{8} \, dx = \frac{x}{8} - \frac{1}{8} \int \cos 4x \, dx.$$

$$\boxed{\int \sin^2 x \cos^2 x \, dx = \frac{x}{8} - \frac{\sin 4x}{32} + C}.$$

THANK YOU!

